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Gamification and personality: Exploring the impact on consumer engagement and purchase intention

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ABSTRACT

The swift progress of digital innovations has prompted companies to more frequently incorporate game-like features to boost customer involvement. This study explores how game-like components, such as badges, points, and leaderboards, influence personality characteristics and their effect on customer involvement and intentions to buy online. A conceptual framework was created to investigate these connections. Through a survey approach, information from 412 people who shop online was gathered and examined using Partial Least Squares Structural Equation Modeling (PLS-SEM). The findings indicate that game-like features have a positive effect on all five main personality dimensions—conscientiousness, openness, extraversion, neuroticism, and agreeableness. Particularly, agreeableness and extraversion were identified as key factors that bridge the gap between game-like features and both customer involvement and intentions to purchase. This study sheds light on how personality traits can affect the success of game-like features in the online retail sector, highlighting the importance of customized marketing approaches that suit different types of customers. To date, this research is among the first to scientifically examine how individual personality traits influence customer involvement and intentions to buy online in a game-like shopping setting.

1. Introduction

"All work and no play makes Jack a dull boy". This timeless proverb underscores the necessity of fun and games in our lives, providing a vital escape from daily routines. As human civilization evolves, new paradigms of work and societal interactions emerge, with technology spearheading these transformations. The strategic integration of technology within management sciences has revolutionized customer experiences, especially through digital interventions that continually reshape our interactions with the market. One notable technological intervention at the crossroads of technology and entertainment is gamification. Gamification entails the incorporation of game-like elements into non-game settings to enhance user engagement and motivation [1]. According to [2], gamification can transform routine activities into more engaging and enjoyable experiences. Recognized as a significant trend by the Gartner Hype Cycle [3], gamification has become a breakthrough strategy for marketers aiming to boost consumer motivation and drive positive marketing outcomes. Research indicates that gamified systems can significantly boost user participation and satisfaction by creating an engaging and enjoyable experience [4]. [5] emphasized the importance of digital interventions in engaging consumers and providing businesses with a competitive advantage. This mutually beneficial scenario relies significantly on the understanding and delivery of appropriate content tailored to consumer behaviour, enhancing consumer attitudes towards the business. Research indicates that gamified systems can significantly boost user participation and satisfaction by creating an engaging and enjoyable experience [4]. [6] found that over 70 % of companies listed in the Forbes Global 2000 reported plans to implement gamification strategies for enhancing customer retention and marketing efforts.

Game designs typically involve integrating elements or mechanics, such as badges, points, and leaderboards, into products or services. These elements provide consumers with insights into their performance and simultaneously motivate them to engage more deeply with the activities.[7]. The distinctive feature of gamification asserts its ability to bring fun, entertainment and engagement to any application [8].[9] suggested that every gamification feature is designed to influence users' motivation, ultimately enhancing their experience and increasing their engagement. Gamification aims to create an engaging and game-like experience [10], motivating and guiding users' behaviour to boost

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their engagement with tasks [11]. By employing techniques such as points, scoreboards, and personalized feedback, gamification helps users feel a greater sense of ownership, flow, and purpose in their activities [12]. In consumer-oriented mobile applications and, companies leverage gamification to encourage the use of eCommerce platforms, thereby driving customer loyalty, increasing brand awareness, and enhancing marketing effectiveness [8,13].

The burgeoning interest in gamification, particularly in its capacity to enhance consumer behavior, necessitates a thorough investigation into its nuanced effects. Despite considerable attention, the differential impact of gamification on various demographic segments, including personality traits and gender differences, remains underexplored. Personality traits, notably, have been identified as significant determinants of consumer decisions [14]. Several frameworks, such as the Big Five [15], HEXACO [16], MBTI [17], HEXAD gamer type [18], and Bartle's player types [19], categorize individuals based on consistent behavioural patterns and intrinsic qualities, offering a fertile ground for analysing gamification's effectiveness.

The Big Five personality model, comprising Neuroticism (emotional instability), Agreeableness (altruism), Openness (willingness to experience new things), Conscientiousness (task-focused and orderly behaviour), and Extraversion (sociability and outgoing nature), is one of the most widely utilized frameworks for personality assessment. The Big Five's adaptability across various contexts, including digital and gamified settings, distinguishes it from other personality frameworks that may be more context-specific [20]. This versatility is particularly valuable in the study of consumer engagement, as it allows for the model's application across a broad spectrum of scenarios and user interactions, thereby enhancing our understanding of how intrinsic personality traits influence engagement with gamification elements such as leaderboards, challenges and point systems. Notably, other personality models can be effectively mapped onto the Big Five, demonstrating its comprehensive nature [21].

Specific traits within the Big Five align directly with particular gamification mechanics, facilitating a deeper analysis of user behaviour. For instance, individuals scoring high in Conscientiousness are typically drawn to goal-oriented tasks and structured challenges, reflecting their preference for order and achievement [22]. Conversely, those high in Neuroticism are more likely to respond to reward systems that provide reassurance and validation, highlighting how gamified learning environments can be tailored to personality-driven needs [22].

Empirical research suggests that the diverse personality profiles of individuals shape their reactions to gamified elements, potentially eliciting a spectrum of emotional responses from curiosity and satisfaction to frustration [23]. Gamification harnesses these emotional and cognitive responses to foster consumer engagement and loyalty. The hedonic motivations, such as the enjoyment perceived by users, are particularly pivotal in determining the success of gamification strategies. This insight is critical for businesses seeking to leverage gamification to bolster customer engagement and drive online purchase intentions. However, the specific effect of personality traits on the effectiveness of gamification in online shopping contexts remains poorly understood. Questions such as whether extroverts are more receptive than introverts to gamified shopping experiences and how different personality types' engagement and purchase intentions are influenced by gamification warrant rigorous empirical investigation. This gap signifies a crucial area for academic inquiry, as understanding these dynamics can greatly enhance the strategic deployment of gamification.

To this end, this research proposes to empirically examine the influence of commonly utilized gamified elements—points, badges, and leaderboards—on consumer behaviour across different personality profiles. By delving into how traits like extraversion or neuroticism affect responses to these game mechanics, this study aims to illuminate the nuanced interplay between personality and gamification. This exploration is not merely academic; it holds substantial practical implications, enabling marketers to tailor gamified experiences that resonate more profoundly with diverse consumer segments, thereby optimizing engagement and purchase outcomes.

The structure of the paper is organized in the following manner. Section 2 studies relevant research on theoretical frameworks linking personality attributes to gamification. Section 3 outlines the methodology for the research model, including the design and data collection process. Section 4 highlights the key findings, which provide insights into the interaction of personality traits with gamification. Section 5 of this work summarizes major findings, discusses theoretical and practical implications, and proposes future research topics on the role of gamification in consumer behaviour.

2. Literature review

2.1. Gamification

The concept of play is intrinsically interwoven into human culture and society. Johan Huizinga's "Homo Ludens: A Study of the Play-Element in Culture" (J. [24] provides a foundational understanding of the significance of play in human culture. Huizinga argues that play is a primary condition for the generation of culture and a critical component in the development of civilization. He defines play as a voluntary activity marked by freedom, separation from "real life," and the creation of its own boundaries in time and space. Huizinga's conceptualization of play highlights its intrinsic value and the joy it brings to participants. This concept supports the theoretical framework of gamification, which utilizes the inherent human propensity for play to boost engagement and motivation [8].[25] defined gamification as the "art and science of transforming everyday interactions of customers into games that align with business objectives". By featuring game elements, gamification creates a playful environment that stimulates intrinsic motivation and fosters deeper engagement [26].

Consistent with the literature, research landscape on gamification in the domain of marketing has evolved rapidly in recent years. From brand engagement [27,28] to advertising [29,30], to customer loyalty [31,32] and improved customer experiences [33], gamification has found its place. Huotari and Hamari [26] described gamification as "The process of creating 'gameful experiences' to enable overall value creation." These gameful experiences refer to interactions that incorporate game-like elements to augment user engagement, motivation, and satisfaction (Deterding 2011).

[4] provided evidence supporting the use of gamification to encourage behaviour change in various contexts. The literature on gamification consistently shows that gaming is generally enjoyable, engaging, and intrinsically motivating [34]. The literature consistently demonstrates that gamification boosts engagement and motivation, leading to favourable consumer responses [35,36,37].

2.2. Gamification and consumer engagement

Customer engagement (CE) has been developed as a crucial concept for understanding and fostering the interactions between firms and customers, particularly within virtual contexts. CE is defined as "the level of a customer's behavioural, cognitive, and emotional investment in specific brand interactions"[38]. This concept extends beyond transactional behaviours to include the emotional and psychological connections customers develop with brands [39]. High levels of engagement are linked to increased customer loyalty, satisfaction, and advocacy, which are critical for business success [40]. The digital age has amplified the significance of CE, as interactions between firms and customers increasingly occur online. Virtual environments provide numerous touchpoints for engagement, including social media, websites, and online communities [41]. These platforms facilitate continuous and dynamic interactions, enhancing the potential for deep, sustained engagement [42].

At the heart of gamification, is the capacity to engage users' intrinsic motivations. According to [4], gamified systems can substantially enhance user engagement by rendering activities more enjoyable and rewarding. Engagement is understood as a psychological phenomenon that develops in consumers when they encounter a distinctive experience that fulfils both practical needs and social expectations. The interactive, enjoyable, and user-friendly aspects of gamification maintain customer interest and foster sustained interaction with the brand [43].

Recent studies have significantly advanced our comprehension of how gamification affects consumer engagement, particularly through the lens of digital marketing and customer experiences. Research has demonstrated that gamification effectively enhances customer experience by positively impacting brand engagement, which in turn influences the overall customer experience in online settings. Specifically, gamification strategies such as using, challenges, points, levels and competitions have been found to enrich customer interaction and engagement, thereby facilitating a deeper connection with the brand and enhancing the customer journey from discovery through purchase [44].

2.3. Gamification and PBL

Reward systems in online shopping are pivotal in shaping consumer behaviour and enhancing the overall shopping experience. By offering tangible and intangible incentives, these systems motivate consumers to engage more frequently and deeply with the platform. Rewards such as discounts, loyalty points, and exclusive offers provide immediate gratification, encouraging repeat purchases and fostering brand loyalty [45]. When consumers know they can earn rewards for their actions, they are more likely to spend time exploring products, making purchases, and even recommending the platform to others. It is widely acknowledged that three key elements of gamification—points, badges, and leaderboards—are commonly used in games.[46] refer to these elements as "The PBL Triad." According to [47] and [48], game elements like levels, leaderboards and points have been put to use to a diverse range of non-game contexts. In online shopping, gamification components such as "*Points, Badges, and Leaderboards"* (PBL) are considered significant enablers of behavioural intention (BI) [49]. According to Snipp (2017), companies specifically utilize badges and leaderboards to enhance consumer engagement on online shopping websites or apps. For instance, Nike and Starbucks utilize gamification to enhance user engagement and drive sales. Nike's NikePlus app awards badges for fitness achievements, which users can share on social media, linking these achievements to promotions that encourage purchases. Similarly, Starbucks' app rewards users with "stars" for purchases that can be converted into badges and redeemed for free products, boosting visit frequency and customer loyalty.

Points serve as a straightforward and immediate form of feedback, rewarding users for completing tasks and encouraging continued participation. Accumulating points allows users to monitor their achievements and progress, enhancing their intrinsic motivation and overall satisfaction. Additionally, these points can unlock new levels or content, offering a sense of advancement and introducing continuous challenges [4]. Badges, act as symbolic rewards that acknowledge specific accomplishments or milestones. These visual tokens of achievement not only recognize users' efforts but also serve to enhance their sense of competence and status within a community. Badges can be displayed on user profiles, fostering a sense of pride and encouraging further participation. They also introduce a social element, as users can compare their badges with those of their peers, creating a competitive and motivating environment [50]. Leaderboards infuse a competitive aspect to gamified systems by ranking individuals according to their performance levels [51]. This form of social comparison motivates participants to augment their performance to ascend the rankings and secure superior positions. Leaderboards tap into users' desire for recognition and status, which can significantly boost engagement and motivation. However, it's important to design leaderboards carefully to ensure they foster healthy competition rather than discouraging users who may feel they cannot compete with top performers [46]. These gamified elements have been integrated into educational platforms [52], and health and fitness applications [53] among others. The effectiveness of gamified systems frequently hinges on both the context in which they are applied and their design. When implemented thoughtfully, these elements can significantly enhance user engagement, motivation, and overall experience.

2.4. Gamification and personality

Personality encompasses a collection of stable traits and psychological factors that shape an individual's thoughts, interactions, and responses to diverse situations.[20]. The Big Five personality traits framework, also known as the "Five-Factor Model" (FFM)[54], is a well-researched and broadly accepted tool for understanding personality, offering a robust, empirically validated framework for personality assessment.

While numerous personality models exist, such as *Type A, B, C, and D* Personalities, *HEXACO*, and the *MBTI*, the *FFM* stands out due to its comprehensive taxonomy— Conscientiousness, Openness, Extraversion, Neuroticism, and Agreeableness—and its stable, heritable traits that appear impervious to social influence from peers or parents [55,56]. This model's reliability is underpinned by consistent measurements across different assessment methods, including self-reports, interviews, and observational studies, spanning a wide demographic spectrum. [57;58]. Its scientific rigor and the depth of insight it provides into human behaviour render the Big Five model an indispensable tool in the fields of psychology and behavioural science.

The Five Dimensions of the FFM.

- **Openness to Experience (Openness):**

This trait includes qualities such as creativity, intellectual curiosity, imagination, perceptiveness, and artistic.[59].

- **Conscientiousness:**

This trait reflects an individual's inclination toward towards being timely, decisive, organized, consistent, and dependable.[59].

- **Extraversion:**

This trait indicates a person's tendency toward talkative, assertiveness, outgoing, adventurousness, sociability and active engagement. [59].

- **Agreeableness:**

This dimension includes modesty, compassion, humility, affection, trust and generosity.[59].

- **Neuroticism (Emotional stability):**

Neuroticism is characterized by moodiness, anxiety, irritability, sadness and emotional instability, often influencing stress handling and personal relationships [59].

Understanding these personality dimensions is crucial when assessing the bearings of gamification elements on user engagement. The FFM provides a parsimonious and widely accepted framework for studying personality traits, often represented by the acronym OCEAN, encompassing Openness, Agreeableness, Extraversion, Conscientiousness, and Neuroticism [60]. This study relies on the FFM to explore how gamified elements can be tailored to different personality traits to enhance consumer engagement in various contexts, including online shopping. By leveraging insights from the FFM, businesses can better design gamification strategies that resonate with diverse user personalities, ultimately driving higher engagement, satisfaction, and loyalty.

2.5. Gamification and purchase intention

Purchase intention is a crucial concept in marketing and consumer behaviour research that represents the probability of a consumer choosing to purchase a service or product. This intention is shaped by various factors, including consumer attitudes towards the product, perceptions of product quality, pricing perceptions, and brand loyalty. [61]. Gamification has increasingly become an integral strategy in e-commerce towards enhancing engagement and driving online purchase intentions. Incorporating game mechanics into business platforms not only enriches user interaction but also fosters a fun and rewarding experience that can drive sales. Research supports the idea that gamification elements like points, leaderboards, and interactive challenges can significantly impact the purchasing behaviour of consumers. These elements, often designed to increase user enjoyment and engagement, tap into intrinsic motivations that can direct towards a heightened purchase intention [62,63].

In e-commerce, gamification significantly enhances consumer satisfaction and impacts purchasing decision where, key gamification elements such as rewards, autonomy, and engaging challenges play essential roles in increasing consumer enjoyment and motivating purchase behaviour [64,65]. These elements intend to meet the consumers' psychological needs, making shopping experiences both engaging and rewarding [66].

Research has shown that consumers with previous experience in gaming are more inclined to purchase products that utilize gamified elements. This suggests that familiarity with game mechanics can translate into positive shopping behaviours, with intrinsic and extrinsic motivations from games impacting purchasing decisions [7]. For example, Alibaba has successfully integrated gamification strategies that leverage both social responsibilities and consumer engagement to stimulate sales.

Furthermore, gamified challenges such as unlocking specific products enhance the shopping experience by offering tangible goals and rewards, which have been shown to increase the likelihood of purchase [67]. The findings support that consumers are more likely to buy when online commerce activities include entertaining elements like games [68].

To capitalize on these insights, researchers such as [69] recommend the adoption of gamification as a strategized approach to persuade consumer purchase intentions and drive sales. This comprehensive integration of gamification within e-commerce highlights its effectiveness in not only attracting consumers but also in converting their engagement into actual purchases.

3. Material and methods

3.1. Research Instrument

The research framework integrates elements of gamification, personality traits, customer engagement, and purchase intention. All measures were adopted from established sources: the Gamification Triad was measured as a second order construct, the items of whom were adopted from existing literature (Table 1), Personality Traits were

Table 1
Research measurement scale.

Constructs		Codes	Items	References
Gamification Triad	Points	PTS1	To what extent do earning points positively influence your shopping behaviour?	[46,78,79,80]
		PTS2	How important are reward points to you when deciding which online shopping platform to use?	
		PTS3	Earning points makes me more likely to engage with the online platform.	
		PTS4	I feel motivated to reach higher point milestones for greater rewards.	
		PTS5	Points make the shopping experience more enjoyable for me.	
	Badges	BD1	How motivated are you to earn or collect badges during your online shopping experiences	[46,78,81,82]
		BD2	Earning badges gives me a sense of achievement	
		BD3	Badges make me feel more connected to the community or brand.	
		BD4	The visibility of badges to other users enhances my engagement with the platform	
	Leaderboards	LB1	Leaderboards create a sense of competition that enhances my motivation.	[46,78,80,83]
		LB2	I am motivated to improve my ranking on leaderboards by purchasing more/engaging more.	
		LB3	Leaderboards encourage me to spend more time on the platform to improve my ranking.	

(*continued on next page*)

Table 1 (*continued*)

Constructs	Codes	Items	References
	LB4	Seeing my name on a leaderboard motivates me to engage more with the platform.	
Openness to experience	OP1	I get excited by new ideas.	[70]
	OP2	I enjoy thinking about things.	
	OP3	I enjoy hearing new ideas.	
	OP4	I enjoy looking for a deeper meaning.	
	OP5	I have a vivid imagination.	
Agreeableness	AG1	I sympathize with others' feelings.	[70]
	AG2	I am concerned about others.	
	AG3	I respect others.	
	AG4	I believe that others have good intentions.	
	AG5	I trust what people say to me.	
Conscientiousness	CN1	I carry out my plans.	[70]
	CN2	I pay attention to detail.	
	CN3	I am always prepared.	
	CN4	I make plans and stick to them.	
	CN5	I am exact in my work.	
Neuroticism	NU1	I get stressed out easily.	[70]
	NU2	I worry about things.	
	NU3	I fear the worst.	
	NU4	I am filled with doubts.	
	NU5	I panic easily	
Extraversion	EX1	I talk to a lot of different people at parties.	[70]
	EX2	I feel comfortable around people.	
	EX3	I start conversations	
	EX4	I make friends easily.	
	EX5	I do not mind being the centre of attention.	

Table 1 (*continued*)

Constructs	Codes	Items	References
Customer Engagement	CE1	I recommended the gamified app/website to others.	[39,44,71,72,73,74]
	CE2	I participate in the challenges or competitions offered by the app/website.	
	CE3	Interacting with the gamified features (badges, points, leaderboards) make me feel more connected to the app/ website.	
	CE4	I feel a sense of joy or fun when using the gamified app/ website.	
	CE5	I find myself thinking about the gamified app/website.	
Purchase Intention	PI1	Given my experiences with the gamified features, I intend to shop online through this platform in the future.	[75,76,77]
	PI2	I plan to increase my shopping activities on this online platform because of its engaging gamified features.	
	PI3	The gamified elements of this website/ app make me intend to continue buying online here in the future.	
	PI4	I will recommend this gamified online store to others due to the positive experiences I've had.	
	PI5	Because of the enjoyable gamified shopping	

(*continued on next page*)

Table 1 (*continued*)

Constructs	Codes	Items	References
		experience, I will encourage my friends and family to shop on this online platform.	

measured using instruments adopted from [70], Customer Engagement scales were drawn existing empirical literature [44,71,72,73,74,39], and Purchase Intention was assessed using the items suggested by [75,76,77]. These adapted items are slightly modified to fit the present study. The survey employed a five-point Likert scale to gauge responses (where 1 denotes "strongly disagree" and 5 denotes "strongly agree").

The questionnaire begins with a concise introduction that delineates the study objectives, setting the stage for the respondents. The initial section of the questionnaire introduces the concept of gamification. To ensure a thorough understanding, a 38-second YouTube video explaining gamification is provided to provide a visual and definitive explanation. This approach aimed to ensure respondents had a clear understanding before proceeding with the survey questions. Subsequently, the survey queried respondents on their understanding of gamification within the online shopping context. Following this, the questionnaire explored the elements of Points, Badges and Leaderboards (PBL), assessed individual personality traits, and constructs related to customer engagement and purchase intentions. The data was collected between the months of April and August 2024.

3.2. Sample and data collection

In total, 500 questionnaires were distributed via mail and in-person visits in June 2024. The survey initially received 440 responses. After excluding responses that were incomplete or improperly filled out, the analysis proceeded with 412 (responses response rate = 88 %). A priori power analysis was conducted using G*Power [84] to determine the required sample size for a multiple regression analysis with 9 predictors. The analysis assumed a medium effect size ($f^2 = 0.15$), an alpha level of 0.05, and a desired power of 0.80. This setting is widely recommended for social and business science research, as [85] suggested. The analysis determined that at least 114 participants were necessary to ensure the study had sufficient power to detect significant effects. To bolster the robustness and enhance the generalizability of the findings, the survey questionnaires were distributed to a larger sample comprising of 412 participants (Table 2).

Table 2
Descriptive of sample characteristics.

Measure	Item	Count
Gender	Male	198
	Female	214
Age	Below 18	7
	18–30	363
	31–40	29
	41–60	12
	Above 60	1
Education Level	High School	16
	Diploma	5
	Bachelor's Degree	289
	Master's Degree	59
	Doctorate	43

3.3. Hypotheses development

Existing studies and literature in the online shopping context have investigated the effects of personality traits on consumer behaviour, often tailoring hypotheses to their specific study needs.[86] found a positive association between online shopping and openness, and extraversion suggesting that these traits enhance consumers' willingness to engage in digital shopping environments. In contrast, [87] identified that agreeableness, extraversion and conscientiousness positively influence purchase willingness in physical stores but do not extend these effects to online shopping contexts [88]. further examined the impact of personality traits on consumer attitudes to advertisements, confirming that individuals high on conscientious, extraversion, neuroticism and openness positively affect attitudes toward ads and brands, and bolster purchase intentions.

While research on the interaction between gamified elements and personality traits in online shopping is limited, significant work has been conducted within gamified learning environments to understand how personality characteristics influence user engagement. For instance, a study by [89] revealed that participants with high agreeableness tend to have a favourable view of the points system in gamification, indicating a preference for this feature. Conversely, conscientiousness had a negative impact, as students with lower conscientiousness enjoyed points more than their highly conscientious peers, which deviates from the expected goal-oriented nature of conscientious individuals. Neuroticism also negatively affected the enjoyment of points aligning [22]. Openness showed no significant effect on perceptions of points, badges, and leaderboards, partially matching findings by [90], although study by [91] revealed that openness could influence the perceptivity of points. Extraversion did not significantly impact the perception of gamified elements like points, badges, or leaderboards [92–93].

Drawing from prior research, it is essential to delve into the complex interactions between gamification elements and the Big Five personality traits within the specific realm of this study. Consequently, this research endeavours to explore how these gamification mechanisms relate to various personality dimensions. Consequently, the authors propose the following hypotheses to guide this investigation

H1 Gamification triad positively affects Extraversion.

H2 Gamification triad positively affects Openness to experience.

H3 Gamification triad positively affects Neuroticism.

H4 Gamification triad positively affects Conscientiousness.

H5 Gamification triad positively affects Agreeableness.

H6 Consumer Engagement positively affects Purchase intention.

In the second part of our investigation, we delve deeper into the overarching impact of personality characteristics as mediators in the relationship between gamification elements—specifically the triad of points, badges, and leaderboards—and consumer engagement, which ultimately leads to purchase intention in online retail settings. Existing literature suggests that personality traits can significantly influence how consumers are engaged and exhibit positive shopping behaviours [87,94]. By acting as a mediator, personality traits like extraversion, conscientiousness, and agreeableness can either amplify or diminish the impact of gamification elements on consumer engagement. For instance, extroverted individuals may find leaderboards particularly motivating, enhancing their engagement and driving their intention to purchase, whereas conscientious individuals may respond differently, focusing on elements that align with their structured and goal-oriented nature [93]. Understanding this mediation effect is critical as it provides insights into how tailored gamification strategies can be used to enhance consumer experiences and drive purchase behaviours. Building on this foundation, the authors propose the following hypotheses:

H7 Extraversion positively mediates the relationship between gamification elements (points, badges, and leaderboards) and consumer engagement, leading to increased purchase intention in online shopping.

H8 Openness to experience positively mediates the relationship between gamification elements (points, badges, and leaderboards) and consumer engagement, leading to increased purchase intention in online shopping.

H9 Neuroticism positively mediates the relationship between gamification elements (points, badges, and leaderboards) and consumer engagement, leading to increased purchase intention in online shopping.

***H*10** Conscientiousness positively mediates the relationship between gamification elements (points, badges, and leaderboards) and consumer engagement, leading to increased purchase intention in online shopping.

***H*11** Agreeableness positively mediates the relationship between gamification elements (points, badges, and leaderboards) and consumer engagement, leading to increased purchase intention in online shopping.

Fig. 1 shows the conceptual framework, based on the hypothesized relationships.

3.4. Data analysis and procedure

SEM (Structural Equation Modeling) is typically used for explaining multiple statistical relationships simultaneously, providing both model validation and visualization and is generally confirmatory, rather than exploratory [95]. The authors used SmartPLS (Partial Least Squares), Ringle et al. [96] for the quantitative data analysis. PLS-SEM is particularly suitable for analysing complex models, focusing on prediction to enhance external validity, accommodating data that do not meet normal distribution assumptions, incorporating formative constructs, and utilizing higher-order constructs to deepen the understanding of theoretical models. [97,98]. It analyses the given data in two phases- Measurement and Structural model. The Measurement Model ensures variables correctly represent constructs through validity tests, while the Structural Model evaluates hypothesized construct relationships and predictive accuracy using distinct metrics.

In terms of model structure, the Gamification Triad is conceptualized as a second-order reflective formative construct in the Partial Least Squares Structural Equation Modelling (PLS-SEM) framework, comprising points, leaderboards, and badges—collectively forming the Triad. This configuration is often referred to as a Type-II higher-order construct. In such formative models, the Lower Order Constructs (LOCs) do not necessarily share a common cause but collectively constitute a broader concept that fully mediates the impact on subsequent endogenous variables. This structure allows for a nuanced understanding of how gamification elements interact with personality traits to influence customer engagement and ultimately drive purchase intentions.

[99] emphasize that, the decision regarding the measurement of a construct is crucial for performing the analysis in PLS SEM, as incorrect measurement model specification can lead to flawed evaluations of relationships in PLS path modelling. The relationship between indicators and an unobservable construct can be defined as either reflective or formative. Reflective measurement, the more commonly applied approach, considers indicators as manifestations of the latent construct, meaning they are influenced by changes in the construct itself [100,101]. In contrast, formative measurement assumes that indicators shape or cause the latent construct, so variations in these indicators directly impact the constructs [102]. Furthermore, traditional methods used for assessing the validity and reliability of reflective constructs do not apply to formative constructs, necessitating distinct evaluation techniques [99].

The gamification triad draws upon a formative measurement approach, wherein the first-order constructs define the attributes of the second-order construct. The removal of any first-order construct would change the second-level construct's conceptual domain, and any modifications to the first-order constructs would directly affect the overall construct. Furthermore, the fact that the first-order constructs are not interchangeable further supports the measurement's formative nature. *Type II models* are those in which the second-order construct (Gamification Triad) is assessed formatively, while the first-order constructs are measured reflectively [103,102]. The *reflective-formative* model was chosen because the measures used to assess the first-order constructs are unidimensional, meaning each first-order construct is represented by indicators that reflect a single underlying dimension or trait.

3.4.1. Operationalizing second order constructs

For assessing the second order constructs, a two-staged approach was followed. For the first stage, this approach involves estimating the scores of the first-order constructs. During the second stage, the scores derived from the first stage serve as indicators for the second-order construct, while also estimating the path coefficients between the other constructs [104,105].

3.5. Common method bias

An assessment of common method bias in the study was conducted using Harman's single-factor test [106]. The test results revealed that a single factor explained 31.353 % of the overall variance, considerably below the 50 % criterion. Therefore, the findings indicate that the presence of common method bias is not a significant issue in our study.

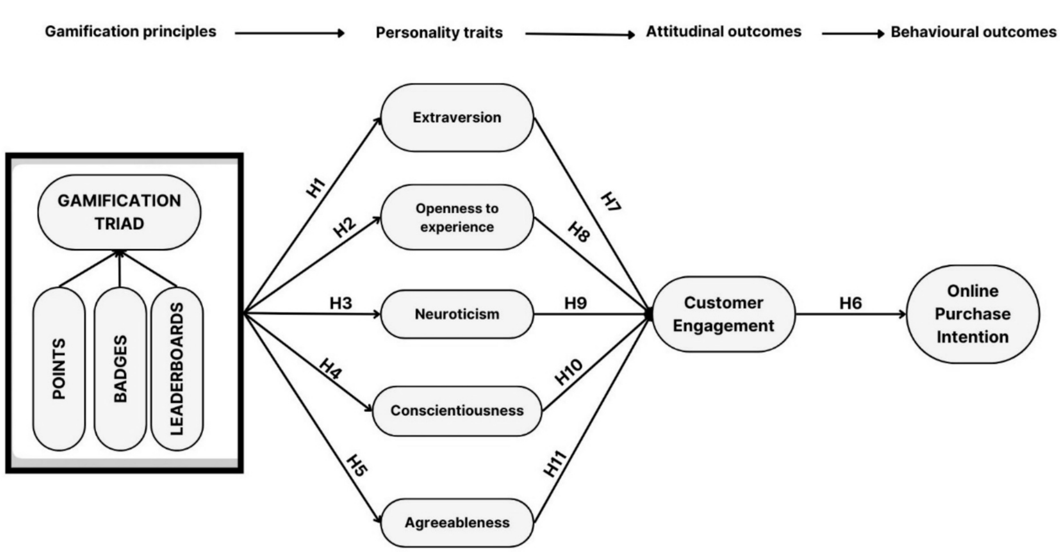

Fig. 1. Theoretical Framework.

4. Results and findings

In Partial Least Squares Structural Equation Modeling (PLS-SEM), analysis is generally segmented into two principal phases: the measurement model and the structural model. Initially, the quality of the measurement model is assessed through a series of reliability and validity tests. Following this, the theoretical hypotheses are evaluated using the structural model. This structured two-step approach allows for a thorough examination of both the constructs and their interrelationships within the model.

4.1. Measurement model

4.1.1. Reliability and validity

The reliability of the constructs were assessed through Composite Reliability (CR) and Cronbach's Alpha and indices. In line with [107], the study examined factor loadings and squared multiple correlations between the items and their corresponding constructs to validate the

Table 3
Reliability and Convergent Validity measures.

Constructs	Items	Loadings/Weights	Cronbach's alpha	Composite reliability	Composite reliability	Average variance extracted	VIF
Points	PTS1	0.749	0.875	0.884	0.909	0.667	1.832
	PTS2	0.756					1.767
	PTS3	0.861					2.572
	PTS4	0.857					2.846
	PTS5	0.853					2.361
Badges	BD1	0.838	0.898	0.898	0.929	0.765	2.069
	BD2	0.887					2.806
	BD3	0.905					3.309
	BD4	0.868					2.518
Leaderboards	LB1	0.877	0.926	0.927	0.948	0.819	2.647
	LB2	0.918					3.699
	LB3	0.922					4.276
	LB4	0.903					3.204
Openness	OP1	0.869	0.879	0.946	0.909	0.667	2.092
	OP2	0.849					2.355
	OP3	0.871					3.018
	OP4	0.776					2.261
	OP5	0.708					1.715
Agreeableness	AG1	0.632	0.791	0.841	0.840	0.517	2.066
	AG2	0.736					2.308
	AG3	0.579					1.647
	AG4	0.835					1.747
	AG5	0.782					1.572
Conscientiousness	CN1	0.760	0.821	0.832	0.874	0.581	1.626
	CN2	0.729					1.705
	CN3	0.774					1.685
	CN4	0.805					1.828
	CN5	0.740					1.674
Neuroticism	NU1	0.914	0.852	1.309	0.886	0.662	2.567
	NU2	0.791					2.326
	NU3	0.752					2.087
	NU4	0.713					1.835
	NU5	0.884					2.464
Extroversion	EX1	0.775	0.871	0.879	0.907	0.662	1.830
	EX2	0.799					1.976
	EX3	0.872					2.646
	EX4	0.869					2.488
	EX5	0.745					1.688
Customer engagement	CE1	0.814	0.885	0.890	0.915	0.683	2.326
	CE2	0.826					2.671
	CE3	0.830					2.397
	CE4	0.841					2.444
	CE5	0.822					2.188
Purchase intention	PI1	0.867	0.936	0.937	0.951	0.797	2.754
	PI2	0.897					3.924
	PI3	0.923					4.728
	PI4	0.888					3.663
	PI5	0.887					3.700

measures further. Table 3 depicts the reliability and convergent validity of constructs. Items with factor loadings below 0.4 are typically recommended for removal [108], yet a threshold of 0.60 is often required to affirm convergent validity [109]. In this analysis, no items exhibited loadings below 0.50, as indicated by [110], which supported the decision to retain all items for subsequent analysis. Additionally, the results indicated that Cronbach Alpha values for all constructs surpassed the benchmark of 0.7, as proposed by [111], thereby confirming their reliability.

The reliability of the measurement model was further validated through an assessment of Composite Reliability (CR), which [112] identified as a more rigorous reliability measure. The analysis revealed that all CR values exceeded 0.8, meeting the criteria for strong reliability. The elevated levels of both Cronbach's Alpha and CR affirm the robust reliability of the measurement model.

Furthermore, the average variance extracted exceeds the threshold of 0.5, establishing convergent validity. Discriminant validity was established through HTMT ratios and Fornell Larcker criterion [113]. The observed values fell below the threshold of 1.00 [114], indicating satisfactory discriminant validity. Multicollinearity was evaluated by examining the Variance Inflation Factor (VIF) for each indicator. The analysis confirmed that all VIF values were below the threshold of 5, indicating no concerns of multi-collinearity.

4.1.2. Higher order construct (Validating triad)

The gamification triad is a second-order formative construct measured by three lower-order reflective constructs (Badges, Leaderboards and Points). To establish the reliability and validity of this second-order construct, path coefficients, VIF and outer weights were assessed. VIF values were less than 5. As suggested by [115], VIF values less than 5 possess no multi-collinearity issues. The statistical significance was established using outer weights and outer loadings [98]. Apart from the Leaderboards, the outer weights for the other two indicators were deemed significant. For this, outer loadings for all three constructs were checked. The outer loadings for each indicator associated with Triad were found to be significant. Consequently, this affirmed the validity of the higher-order construct. The HTMT ratio and Fornell and Larcker criterion [113] are also assessed for establishing the discriminant validity (Tables 4 and 5). According to the Fornell-Larcker criterion, the square root of the Average Variance Extracted (AVE) by a construct should exceed the correlations between that construct and any other construct in the model. This means the diagonal value, which represents the square root of AVE, should be larger than all other values in the corresponding column and row of the correlation matrix. Meeting this criterion along with ensuring that the Heterotrait-Monotrait (HTMT) ratio is less than 0.90, as suggested by [116], confirms the discriminant validity of the constructs in the dataset. This analysis confirms that each construct is distinct and represents phenomena not accounted for by other constructs in the framework.

4.2. Structural model

Following the establishment of validity and reliability of the measurement model, it is essential to analyse the estimates of the structural model. This analysis is crucial for evaluating the proposed hypothetical relationships among the constructs within the conceptual framework [117,118]. The model explanatory power is assessed through values of R^2 and Q^2. The values of determination of coefficients (R^2) for each endogenous construct is presented and analysed. Table 6 shows values of Agreeableness 0.137, Conscientiousness 0.062, Extraversion 0.076, Neuroticism 0.025, Openness 0.138, Customer Engagement 0.139, and Purchase Intention 0.619. The results show that values of R^2 are between 0 & 1. According to [119], the R-square values are weak to substantial. [120] also recommended the values falling within the range of weak to substantial.

The predictive relevance is established through Q^2 value [121]. A Q^2 value above zero shows that the model has predictive relevance. Agreeableness 0.126, Conscientiousness 0.050, Extraversion 0.067, Neuroticism 0.017, Openness 0.129, Customer Engagement 0.207, and Purchase Intention 0.176. The range of values suggest that the degree of predictive relevance is between weak to strong [119].

Bootstrapping was employed to evaluate the significance of the hypothesized relationships, following the methodology recommended by [122] Sarstedt et al. (2017). Consistent with [122,123] suggestion, a bootstrapping sample size of 10,000 was utilized for this analysis.

The direct relationships of the hypothesized relationships are presented in Table 7. The results highlight a significant positive impact of the gamification triad (points, badges and leaderboards) on each of the five personality traits. Triad → Agreeableness (H1: $\beta = 0.379$, $t = 8.538$, $p = 0.000$); Triad → Conscientiousness (H2: $\beta = 0.254$, $t = 4.859$, $p = 0.000$); Triad → Extraversion (H1: $\beta = 0.281$, $t = 5.088$, $p = 0.000$); Triad → Neuroticism (H1: $\beta = 0.156$, $t = 2.107$, $p = 0.035$); Triad → Openness (H1: $\beta = 0.369$, $t = 7.688$, $p = 0.000$) and Customer engagement → (H6: $\beta = 0.787$, $t = 43.301$, $p = 0.000$.) Based on these outcomes, hypotheses H1, H2, H3, H4, H5 and H6 were confirmed for the overall sample.

4.3. Mediation analysis

After evaluating the direct relationships, a mediation analysis was conducted to further explore the indirect effects within the model. Overall, the personality traits of Agreeableness and Extraversion were found to have a significant positive impact- Triad → Extraversion → Customer Engagement ($\beta = 0.056$, $t = 2.207$, $p = 0.027$) and Triad → Agreeableness → Customer Engagement ($\beta = 0.060$, $t = 2.156$, $p = 0.03$) (Table 8). The same was validated through the overall indirect effect following the relation Triad –> Agree –> Customer engagement –> Purchase intention ($\beta = 0.047$, $t = 2.127$, $p = 0.033$), TRIAD –> Extraversion –> Customer engagement –> Purchase intention ($\beta = 0.044$, $t = 2.195$, $p = 0.028$). The indirect effects involving the other three personality traits did not demonstrate statistical significance. The mediating effects were further corroborated by analysing the

Table 4
Heterotrait Monotrait Matrix (HTMT).

	AG	BD	CN	CE	EX	LB	NU	OP	PI	PTS
AG										
BD	0.381									
CN	0.699	0.271								
CE	0.292	0.760	0.267							
EX	0.400	0.328	0.472	0.338						
LB	0.307	0.875	0.219	0.716	0.248					
NU	0.364	0.149	0.149	0.104	0.058	0.118				
OP	0.707	0.336	0.628	0.264	0.403	0.318	0.120			
PI	0.339	0.780	0.222	0.853	0.266	0.747	0.107	0.274		
PTS	0.375	0.819	0.270	0.712	0.257	0.730	0.118	0.389	0.732	

Table 5
Fornell-Larcker criterion.

	AG	BD	CN	CE	EX	LB	NU	OP	PI	PTS
AG	**0.719**									
BD	0.361	**0.875**								
CN	0.507	0.238	**0.762**							
CE	0.297	0.687	0.233	**0.827**						
EX	0.350	0.291	0.398	0.295	**0.813**					
LB	0.296	0.799	0.197	0.660	0.226	**0.905**				
NU	0.308	0.164	0.114	0.115	−0.004	0.133	**0.814**			
OP	0.513	0.321	0.503	0.256	0.359	0.317	0.100	**0.817**		
PI	0.351	0.715	0.200	0.787	0.241	0.696	0.117	0.267	**0.893**	
PTS	0.345	0.732	0.235	0.635	0.227	0.663	0.127	0.370	0.665	**0.817**

Table 6
Explanatory Tests.

	R-square	Q-square
Agreeableness	0.137	0.127
Conscientiousness	0.062	0.050
Extraversion	0.076	0.067
Neuroticism	0.025	0.016
Openness	0.138	0.121
Customer Engagement	0.139	0.204
Purchase Intention	0.619	0.173

Table 7
Hypothesis Testing (Direct Effects).

Hypotheses	β	T statistics	P values	Decision
Triad –> Agreeableness	0.379	8.535	0.000	Supported
Triad –> Conscientiousness	0.254	4.813	0.000	Supported
Triad –> Extraversion	0.281	5.168	0.000	Supported
Triad –> Neuroticism	0.158	2.544	0.011	Supported
Triad –> Openness	0.369	7.774	0.000	Supported
Customer engagement –> Purchase Intention	0.787	42.216	0.000	Supported

Table 8
Hypothesis Testing (Mediation Effects).

Hypotheses	β	T statistics (|O/STDEV|)	P values	Decision
Triad –> Conscientious –> Customer Engagement –> Purchase Intention	0.005	0.37	0.711	Not supported
Triad –> Agree –> Customer Engagement –> Purchase Intention	0.045	2.024	0.043	Supported
Triad –> Extra –> Customer Engagement –> Purchase Intention	0.045	2.224	0.026	Supported
Triad –> Open –> Customer Engagement –> Purchase Intention	0.025	1.215	0.225	Not supported
Triad –> Neuro –> Customer Engagement –> Purchase Intention	0.007	0.869	0.385	Not supported

confidence intervals (CIs) for the indirect effects. These intervals at both the lower and upper levels did not include zero, confirming the statistical significance of the mediation. (Fig. 2).

5. Discussion

This study presents a novel investigation into the interaction effects of personality traits within a gamified online shopping environment, addressing a significant gap in the existing literature. While gamification has been extensively studied in the contexts of learning and education, its application in online retail settings, specifically through the lens of personality traits, remains underexplored. The findings of this study reveal that gamification elements—namely points, badges, and leaderboards—positively influence consumer engagement and purchase intention, particularly for individuals high in agreeableness and extraversion.

Our hypotheses posited that gamified elements would positively affect all Big Five personality traits and these traits would mediate the relationship between gamification and consumer engagement, leading to increased purchase intention. The results largely support these propositions, particularly highlighting the significant mediating effects of agreeableness and extraversion. The positive mediation pathways—Triad → Extraversion → Customer Engagement → Purchase Intention ($\beta = 0.056$, $t = 2.207$, $p = 0.027$) and Triad → Agreeableness → Customer Engagement → Purchase Intention ($\beta = 0.060$, $t = 2.156$, $p = 0.03$)—underscore the pronounced impact of these traits on consumer engagement and purchasing behaviour in gamified environments. Further validation through confidence interval analysis confirmed the robustness of these mediating effects, with intervals not including zero, thus reinforcing the statistical significance of the mediation.

The outcomes of this study affirm the vital role of gamification in enhancing consumer engagement and purchase intentions, particularly when aligned with the Big Five personality traits. Our investigation revealed that gamification elements effectively amplify consumer involvement and can significantly influence purchasing decisions, echoing the sentiments of prior research that highlights the impact of such mechanisms in digital environments [10]. Significantly, the study identified that the traits of agreeableness and extraversion significantly influence the interaction between gamification and consumer outcomes, indicating that individuals with these traits tend to be more receptive to gamified elements. This receptivity is likely attributed to their natural sociability and sensitivity to social cues. Such insights are vital for marketers who are designing targeted strategies that leverage these personality characteristics, thereby boosting the effectiveness of campaigns designed for these specific personality profiles.

The study's findings align with the Homo Ludens concept, suggesting that humans inherently engage in and value playful activities across various domains, including digital and commercial environments [124]. The significant influence of gamification on all Big Five traits supports this assertion, indicating that gamified elements appeal broadly to different personality types, fostering engagement and enhancing the overall shopping experience. However, the differential impact on personality traits is particularly noteworthy. Agreeableness and extraversion demonstrated the most substantial mediating effects, suggesting that individuals who are cooperative, responsive, energetic, and socially interactive are more positively influenced by gamified elements. These traits amplify the impact of gamification on engagement and purchase behaviours, making such individuals more likely to interact with gamified systems, stay involved, and ultimately make purchases. For example, extraverted consumers may find leaderboards motivating due

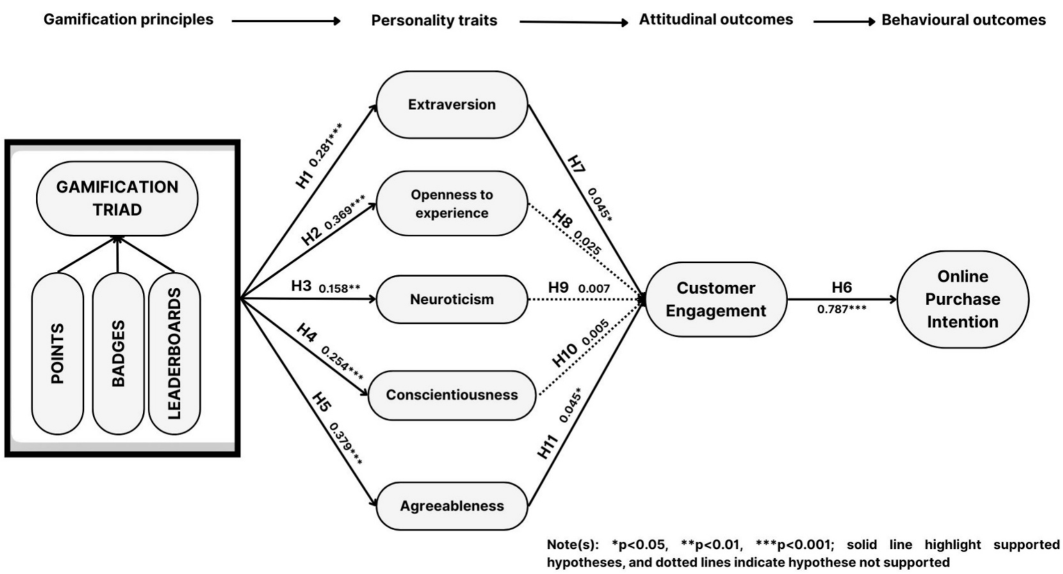

Fig. 2. Results of Structural Model.

to their competitive and social nature, while agreeable consumers are drawn to elements that foster community and cooperation. Conversely, the study found no statistically significant mediating effects for openness, conscientiousness, and neuroticism in the examined context. The absence of significant mediation suggests that gamification strategies in online shopping may need to be tailored differently for these personality types, potentially focusing on specific gamified elements that align more closely with their intrinsic motivations.

5.1. Implications for theory and practice

The results augment our understanding of the crucial interplay between personality traits and gamification in online retail, extending implications for both theory and practice. Theoretically, the study enhances our comprehension of how personality traits influence consumer reactions to gamified elements- potently linking gamification with personality psychology within the online shopping framework. It underscores the significance of considering personality traits when designing gamified consumer experiences, as certain traits can enhance or reduce the effectiveness of these elements. This exploration opens new avenues for exploring how diverse personality traits influence the effectiveness of gamified strategies, significantly transforming consumer behaviour models.

From a practical standpoint, marketers and game designers should leverage these insights to develop more personalized gamification strategies. For instance, platforms can incorporate adaptive gamified systems that adjust to the user's personality profile, enhancing engagement and encouraging positive shopping behaviours. For highly extraverted and agreeable consumers, incorporating social and community-driven elements such as leaderboards, cooperative challenges, or community badges could significantly boost engagement and purchase intention. Conversely, for traits like conscientiousness and openness, gamification strategies might need to incorporate goal-setting or exploratory elements to better align with these consumers' preferences.

Additionally, gamification and its proliferation into online retail invites- broader applications across various sectors, highlighting the effective enhancement in consumer interaction, gamification offers, beyond the traditional educational contexts. This integration across industries implies that gamified elements could be tailored to platforms, for enhancing engagement and satisfaction for users.

Developers should also consider to appraise ethical implications, as the gamified strategies become more personalized. The call for effective guidelines to ensure that such strategies are implemented ethically in a strategized manner, such that they respect user privacy is paramount, especially as systems become more adept at leveraging personal data for behavioural influence. Finally, it is also imperative to account for cultural differences in gamification strategy design to create more sensitive approaches that utilize cultural norms and values, thereby, maximizing engagement.

To sum it up, effective addressal of these implications provides a comprehensive framework for future research and practical applications, ensuring that gamification strategies offer a dual perspective- being ethically sound and seamlessly integrated into the digital consumer landscape.

5.2. Limitations and future directions

While the work provides crucial insights into the interaction between gamification and personality traits in online shopping environments, the research contains inherent limitations. The cross-sectional nature of the study limits the ability to establish causal relationships between gamification elements, personality traits, and consumer behaviours, highlighting the need for studies of a longitudinal nature, that will facilitate in capturing the dynamic nature of these interactions for a prolonged period. Also, the homogeneous nature and size of sample may restrict the generalizability of findings, specifically across varied demographics and cultural contexts. Future research can entail expanding across contexts to allow more robust conclusions. Furthermore, while the reliance on self-reported data provides behavioural insights, subsequent research should incorporate more objective measures to mitigate potential biases. Future studies can employ other types of methodologic advancements, including experiments to better study the consumer behaviour while interacting with gamified platforms.

Future works can attempt to explore the integration of emergent technologies like augmented reality (AR), artificial intelligence (AI), to enhance personalized gamified experiences. These enhancements could provide a more immersive and individualized shopping experiences that offers holistic consumer engagement, aligning closely with consumer personality traits. Further research can entail the interaction of other gamified elements like challenges, levels among others- which can assist in offering a more nuanced view of how different gamified mechanics impact different personality profiles.

Potential works can also delve into exploration of specific industry contexts, such as- healthcare, fashion among others, to investigate the integration of gamification with consumer behaviour in niche environments. The interaction of personality traits with other behavioural factors, such as demographics and social environments, can also provide in-depth insights into consumer engagement, within a gamified setting.

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Rajshree Misara: Investigation, Writing – original draft, Data curation, Visualization, Formal analysis, Writing – review & editing, Validation, Conceptualization. **Divyanshu Verma:** Writing – review & editing. **Saurabh Mishra:** Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

References

[1] F. Conejo, Loyalty 3.0: how to revolutionize customer and employee engagement with big data and gamification, J. Consum. Mark. 31 (1) (2014) 86–87, https://doi.org/10.1108/JCM-09-2013-0700.
[2] Gatautis, R., Banytė, J., Kuvykaitė, R., Virvilaitė, R., Dovalienė, A., Piligrimienė, Ž., Gadeikienė, A., Vitkauskaitė, E., & Tarutė, A. (2021). *The Conceptual Model of Gamification-Based Consumer Engagement in Value Creation* (pp. 99–108). 10.1007/978-3-030-54205-4_5.
[3] Gartner. (2011). Hype Cycle for Emerging Technologies, 2011. In *Gartner*. https://www.gartner.com/en/documents/1754719.
[4] Hamari, J., Koivisto, J., & Sarsa, H. (2014). Does Gamification Work? – A Literature Review of Empirical Studies on Gamification. *2014 47th Hawaii International Conference on System Sciences*, 3025–3034. 10.1109/HICSS.2014.377.
[5] Sen, S., Johnson, A. R., Bhattacharya, C. B., & Wang, J. (2015). *Identification and Attachment in Consumer-Brand Relationships* (pp. 151–174). 10.1108/S1548-643520150000012006.
[6] Park, H. J., & Bae, J. H. (2014). *Study and research of gamification design. 8*, 19–28. 10.14257/ijseia.2014.8.8,03.
[7] J.V. Bittner, J. Shipper, Motivational effects and age differences of gamification in product advertising, J. Consum. Mark. 31 (5) (2014) 391–400, https://doi.org/10.1108/JCM-04-2014-0945.
[8] Deterding, S., Dixon, D., Khaled, R., & Nacke, L. (2011). From game design elements to gamefulness. *Proceedings of the 15th International Academic MindTrek Conference: Envisioning Future Media Environments*, 9–15. 10.1145/2181037.2181040.
[9] Harms, J., Seitz, D., Wimmer, C., Kappel, K., & Grechenig, T. (2015). Low-Cost Gamification of Online Surveys. *Proceedings of the 2015 Annual Symposium on Computer-Human Interaction in Play*, 109–113. 10.1145/2793107.2793146.
[10] M. Sigala, The application and impact of gamification funware on trip planning and experiences: the case of TripAdvisor's funware, Electron. Mark. 25 (3) (2015) 189–209, https://doi.org/10.1007/s12525-014-0179-1.
[11] J. Lee, J. Hammer, Gamification in education: what, how, why bother? Acad. Exch. Q. 15 (2011) 1–5.
[12] J. Pavlus, The game of life, Sci. Am. 303 (2010) 43–44, https://doi.org/10.1038/scientificamerican1210-43.
[13] Sigala, M. (2015a). *Gamification for Crowdsourcing Marketing Practices: Applications and Benefits in Tourism*. 129–146. 10.1007/978-3-319-18341-1_11.
[14] Mowen, J. C. (2000). The 3M Model of Motivation and Personality. *The 3M Model of Motivation and Personality*. 10.1007/978-1-4757-6708-7.
[15] R.R. McCrae, P.T. Costa, Validation of the five-factor model of personality across instruments and observers, J. Pers. Soc. Psychol. 52 (1) (1987) 81–90, https://doi.org/10.1037/0022-3514.52.1.81.
[16] Lee, K., & Ashton, M. C. (2004). Psychometric properties of the HEXACO personality inventory. In *Multivariate Behavioral Research* (Vol. 39, Issue 2, pp. 329–358). Lawrence Erlbaum. 10.1207/s15327906mbr3902_8.
[17] I.B. Myers, *Manual: a guide to the development and use of the Myers-Briggs Type Indicator*, Consulting psychologists press, 1985.
[18] G. Tondello, R. Wehbe, L. Diamond, M. Busch, A. Marczewski, L. Nacke, The Gamification User Types Hexad Scale. (2016), https://doi.org/10.1145/2967934.2968082.
[19] Bartle, R. (1996). *Hearts, clubs, diamonds, spades: Players who suit MUDs*.
[20] Goldberg, L. R. (1992). The development of markers for the Big-Five factor structure. In *Psychological Assessment* (Vol. 4, Issue 1, pp. 26–42). American Psychological Association. 10.1037/1040-3590.4.1.26.
[21] W. Kang, Establishing the associations between the big five personality traits and self-reported number of close friends: a cross-sectional and longitudinal study, Acta Psychol. 239 (January) (2023) 104010, https://doi.org/10.1016/j.actpsy.2023.104010.
[22] Codish, D., & Ravid, G. (2014). Personality based gamification: How different personalities percive gamification. *ECIS 2014 Proceedings - 22nd European Conference on Information Systems*.
[23] W. Ghaban, R. Hendley, How different personalities benefit from gamification, Interact. Comput. 31 (2) (2019) 138–153, https://doi.org/10.1093/iwc/iwz009.
[24] Huizinga, J. (1938). *Homo Ludens: A study of the play element in culture*.
[25] Zichermann, G., & Linder, J. (2010). *Game-based marketing: inspire customer loyalty through rewards, challenges, and contests*. 220.
[26] K. Huotari, J. Hamari, A definition for gamification: anchoring gamification in the service marketing literature, Electron. Mark. 27 (1) (2017) 21–31, https://doi.org/10.1007/s12525-015-0212-z.
[27] Y. Singh, R. Milan, Utilitarian and hedonic values of gamification and their influence on brand engagement, loyalty, Trust and WoM. Entertainment Computing 52 (2025) 100868, https://doi.org/10.1016/J.ENTCOM.2024.100868.
[28] T.H. Tseng, S.H. Hsieh, C.T. Lee, How gamified branded applications drive marketing effectiveness? Market. Intel. Plan. 39 (5) (2021) 633–648, https://doi.org/10.1108/MIP-09-2020-0407.
[29] J. Högberg, P. Shams, E. Wästlund, Gamified in-store mobile marketing: the mixed effect of gamified point-of-purchase advertising, J. Retail. Consum. Serv. 50 (2019) 298–304, https://doi.org/10.1016/j.jretconser.2018.07.004.
[30] S. Sreejesh, T. Ghosh, Y.K. Dwivedi, Moving beyond the content: the role of contextual cues in the effectiveness of gamification of advertising, J. Bus. Res. 132 (2021) 88–101, https://doi.org/10.1016/j.jbusres.2021.04.007.
[31] J.C. Bauer, M. Linzmajer, L. Nagengast, T. Rudolph, E. D'Cruz, Gamifying the digital shopping experience: games without monetary participation incentives increase customer satisfaction and loyalty, J. Serv. Manag. 31 (3) (2020) 563–595, https://doi.org/10.1108/JOSM-10-2018-0347.
[32] J. Hwang, L. Choi, Having fun while receiving rewards?: exploration of gamification in loyalty programs for consumer loyalty, J. Bus. Res. 106 (2020) 365–376, https://doi.org/10.1016/j.jbusres.2019.01.031.
[33] I.K. Disse, M. Olsson, Uncovering the gamified customer experience in the retail environment, Int. J. Retail Distrib. Manag. 51 (7) (2023) 955–971, https://doi.org/10.1108/IJRDM-07-2022-0268.
[34] R.M. Ryan, C.S. Rigby, A. Przybylski, The motivational pull of video games: a self-determination theory approach, Motiv. Emot. 30 (4) (2006) 344–360, https://doi.org/10.1007/s11031-006-9051-8.
[35] S. Jang, P.J. Kitchen, J. Kim, The effects of gamified customer benefits and characteristics on behavioral engagement and purchase: evidence from mobile exercise application uses, J. Bus. Res. 92 (2018) 250–259, https://doi.org/10.1016/J.JBUSRES.2018.07.056.
[36] Lounis, S., Neratzouli, X., & Pramatari, K. (2013). *Can Gamification Increase Consumer Engagement? A Qualitative Approach on a Green Case* (pp. 200–212). 10.1007/978-3-642-37437-1_17.
[37] Teotónio, N., & Reis, J. L. (2018). *The Gamification Systems Application Elements in the Marketing Perspective* (pp. 77–87). 10.1007/978-3-319-77712-2_8.
[38] R.J. Brodie, L.D. Hollebeek, B. Jurić, A. Ilić, Customer engagement, J. Serv. Res. 14 (3) (2011) 252–271, https://doi.org/10.1177/1094670511411703.
[39] S.D. Vivek, S.E. Beatty, R.M. Morgan, Customer engagement: exploring customer relationships beyond purchase, J. Mark. Theory Pract. 20 (2) (2012) 122–146, https://doi.org/10.2753/MTP1069-6679200201.
[40] J. Bowden, The process of customer engagement: a conceptual framework, J. Marketing Theory Pract. 17 (2009) 63–74, https://doi.org/10.2753/MTP1069-6679170105.
[41] R.J. Brodie, A. Ilic, B. Juric, L. Hollebeek, Consumer engagement in a virtual brand community: an exploratory analysis, J. Bus. Res. 66 (1) (2013) 105–114, https://doi.org/10.1016/j.jbusres.2011.07.029.
[42] E. Malthouse, C. Hofacker, Looking back and looking forward with interactive marketing, J. Interact. Mark. 24 (3) (2010) 181–184, https://doi.org/10.1016/j.intmar.2010.04.005.
[43] S. Kamboj, S. Rana, V.A. Drave, Factors driving consumer engagement and intentions with gamification of mobile apps, J. Electronic Commerce Organizations (JECO) 18 (2) (2020) 17–35, https://doi.org/10.4018/JECO.2020040102.
[44] M. Jami Pour, K. Rafiei, M. Khani, A. Sabrirazm, Gamification and customer experience: the mediating role of brand engagement in online grocery retailing, Nankai Bus. Rev. Int. 12 (3) (2021) 340–357, https://doi.org/10.1108/NBRI-07-2020-0041/FULL/PDF.
[45] P. Torres, M. Augusto, C. Neves, Value dimensions of gamification and their influence on brand loyalty and word-of-mouth: relationships and combinations with satisfaction and brand love, Psychol. Mark. 39 (1) (2022) 59–75, https://doi.org/10.1002/mar.21573.
[46] D. Hunter, K. Werbach, For the Win: how game thinking can revolutionize your business, University of Pensylvania, 2012.

[47] J. Koivisto, J. Hamari, The rise of motivational information systems: a review of gamification research, Int. J. Inf. Manag. 45 (2019) 191–210, https://doi.org/10.1016/j.ijinfomgt.2018.10.013.
[48] K. Seaborn, D.I. Fels, Gamification in theory and action: a survey, Int. J. Hum Comput Stud. 74 (2015) 14–31, https://doi.org/10.1016/j.ijhcs.2014.09.006.
[49] Y. Feng, J. Ye, Y. Yu, C. Yang, T. Cui, Gamification artifacts and crowdsourcing participation: examining the mediating role of intrinsic motivations, Comput. Hum. Behav. 81 (2017), https://doi.org/10.1016/j.chb.2017.12.018.
[50] J. Antin, E. Churchill, Badges in Social Media: a Social Psychological Perspective. In *Gamification Workshop Proceedings*, 2011.
[51] Costa, J., Wehbe, R., Robb, J., & Nacke, L. (2013). Time's Up: Studying Leaderboards For Engaging Punctual Behaviour. In *Gamification 2013: Proceedings of the First International Conference on Gameful Design, Research, and Applications*. 10.1145/2583008.2583012.
[52] R. Smiderle, S.J. Rigo, L.B. Marques, P. de Miranda, J.A. Coelho, P.A. Jaques, The impact of gamification on students' learning, engagement and behavior based on their personality traits, Smart Learn. Environ. 7 (1) (2020) 3, https://doi.org/10.1186/s40561-019-0098-x.
[53] V. Cotton, M.S. Patel, Gamification use and design in popular health and fitness mobile applications, Am. J. Health Promot. 33 (3) (2019) 448–451, https://doi.org/10.1177/0890117118790394.
[54] R.R. McCrae, O.P. John, An introduction to the five-factor model and its applications, J. Pers. 60 (2) (1992) 175–215, https://doi.org/10.1111/j.1467-6494.1992.tb00970.x.
[55] Ackerman, R., & Thompson, V. A. (2017). Meta-reasoning: Monitoring and control of thinking and reasoning. In *Trends in Cognitive Sciences* (Vol. 21, Issue 8, pp. 607–617). Elsevier Science. 10.1016/j.tics.2017.05.004.
[56] R. Riemann, A. Angleitner, J. Strelau, Genetic and environmental influences on personality: a study of twins reared together using the self- and peer report NEO-FFI scales, J. Pers. 65 (3) (1997) 449–475, https://doi.org/10.1111/j.1467-6494.1997.tb00324.x.
[57] M. Barrick, M. Mount, J. Strauss, Conscientiousness and performance of sales representatives—test of the mediating effects of goal setting, J. Appl. Psychol. 78 (1993) 715–722, https://doi.org/10.1037/0021-9010.78.5.715.
[58] Jang, K. L., McCrae, R. R., Angleitner, A., Riemann, R., & Livesley, W. J. (1998). Heritability of facet-level traits in a cross-cultural twin sample: Support for a hierarchical model of personality. In *Journal of Personality and Social Psychology* (Vol. 74, Issue 6, pp. 1556–1565). American Psychological Association. 10.1037/0022-3514.74.6.1556.
[59] C. Soto, A. Kronauer, J. Liang, Five-Factor Model of Personality. (2015), https://doi.org/10.1002/9781118521373.wbeaa014.
[60] P.T. Costa, R.R. McCrae, Normal personality assessment in clinical practice: the NEO personality inventory, Psychol. Assess. 4 (1) (1992) 5–13, https://doi.org/10.1037/1040-3590.4.1.5.
[61] I. Ajzen, The theory of planned behavior, Organ. Behav. Hum. Decis. Process. 50 (2) (1991) 179–211, https://doi.org/10.1016/0749-5978(91)90020-T.
[62] R. Mitchell, L. Schuster, H.S. Jin, Gamification and the impact of extrinsic motivation on needs satisfaction: Making work fun? J. Business Res. 106 (2017) (2020) 323–330, https://doi.org/10.1016/j.jbusres.2018.11.022.
[63] I. Poncin, M. Garnier, M.S. Ben Mimoun, T. Leclercq, Smart technologies and shopping experience: are gamification interfaces effective? The case of the Smartstore, Technol. Forecast. Soc. Chang. 124 (2017) 320–331, https://doi.org/10.1016/j.techfore.2017.01.025.
[64] X. Wu, S. Santana, Impact of intrinsic and extrinsic gaming elements on online purchase intention, Front. Psychol. 13 (2022), https://doi.org/10.3389/fpsyg.2022.885619.
[65] Y. Xu, Z. Chen, M.-Y.-P. Peng, M.K. Anser, Enhancing consumer online purchase intention through gamification in China: perspective of cognitive evaluation theory, Front. Psychol. 11 (2020), https://doi.org/10.3389/fpsyg.2020.581200.
[66] J.K. Mullins, R. Sabherwal, Gamification: a cognitive-emotional view, J. Bus. Res. 106 (2018) (2020) 304–314, https://doi.org/10.1016/j.jbusres.2018.09.023.
[67] C. Hildebrand, T. Schlager, A. Herrmann, G. Häubl, Product gamification, Adv. Consum. Res. 42 (2014) 664–665.
[68] X. Feng, Y. Li, X. Lin, Y. Ning, Mobile targeting in industrial marketing: connecting with the right businesses, Ind. Mark. Manag. 86 (2020) 65–76, https://doi.org/10.1016/j.indmarman.2019.06.007.
[69] M. Helmefalk, L. Marcusson, Gamification in a servicescape context: a conceptual framework, Int. J. Int. Market. Advertising 13 (1) (2019) 22, https://doi.org/10.1504/IJIMA.2019.097894.
[70] K.-H. Yoo, U. Gretzel, Influence of personality on travel-related consumer-generated media creation, Comput. Hum. Behav. 27 (2) (2011) 609–621, https://doi.org/10.1016/j.chb.2010.05.002.
[71] M. Abou-Shouk, M. Soliman, The impact of gamification adoption intention on brand awareness and loyalty in tourism: the mediating effect of customer engagement, J. Destin. Mark. Manag. 20 (January) (2021) 100559, https://doi.org/10.1016/j.jdmm.2021.100559.
[72] Dessart, L., Veloutsou, C., & Morgan-thomas, A. (2016). *Capturing consumer engagement : duality , dimensionality and measurement. 1376*(January). 10.1080/0267257X.2015.1130738.
[73] L.D. Hollebeek, M.S. Glynn, R.J. Brodie, ScienceDirect consumer brand engagement in social media: conceptualization, scale development and validatio, J. Interact. Mark. 28 (2) (2014) 149–165, https://doi.org/10.1016/j.intmar.2013.12.002.
[74] A. Pansari, V. Kumar, Customer engagement: the construct, antecedents, and consequences, J. Acad. Mark. Sci. 45 (3) (2017) 294–311, https://doi.org/10.1007/s11747-016-0485-6.
[75] T. Huang, Z. Bao, Y. Li, Why do players purchase in mobile social network games? An examination of customer engagement and of uses and gratifications theory, Program 51 (3) (2017) 259–277, https://doi.org/10.1108/PROG-12-2016-0078.
[76] J.-W. Moon, Y.-G. Kim, Extending the TAM for a World-Wide-Web context, Inf. Manag. 38 (2001) 217–230, https://doi.org/10.1016/S0378-7206(00)00061-6.
[77] B. Suh, I. Han, Effect of trust on customer acceptance of internet banking, Electron. Commer. Res. Appl. 1 (2002) 247–263, https://doi.org/10.1016/S1567-4223(02)00017-0.
[78] M. Aparicio, C.J. Costa, R. Moises, Gamification and reputation: key determinants of e-commerce usage and repurchase intention, Heliyon 7 (3) (2021) e06383, https://doi.org/10.1016/j.heliyon.2021.e06383.
[79] Y. Feng, H. Jonathan Ye, Y. Yu, C. Yang, T. Cui, Gamification artifacts and crowdsourcing participation: examining the mediating role of intrinsic motivations, Comput. Hum. Behav. 81 (2018) 124–136, https://doi.org/10.1016/j.chb.2017.12.018.
[80] A. García-Jurado, P. Castro-González, M. Torres-Jiménez, A.L. Leal-Rodríguez, Evaluating the role of gamification and flow in e-consumers: millennials versus generation X, Kybernetes 48 (6) (2019) 1278–1300, https://doi.org/10.1108/K-07-2018-0350.
[81] T. Harwood, T. Garry, An investigation into gamification as a customer engagement experience environment, J. Serv. Mark. 29 (6/7) (2015) 533–546, https://doi.org/10.1108/JSM-01-2015-0045.
[82] E. Kyewski, N.C. Krämer, To gamify or not to gamify? An experimental field study of the influence of badges on motivation, activity, and performance in an online learning course, Comput. Educ. 118 (2018) 25–37, https://doi.org/10.1016/j.compedu.2017.11.006.
[83] R.N. Landers, K.N. Bauer, R.C. Callan, Computers in human behavior Gami fi cation of task performance with leaderboards: a goal setting experiment, Comput. Hum. Behav. 71 (2017) 508–515, https://doi.org/10.1016/j.chb.2015.08.008.
[84] F. Faul, E. Erdfelder, A.G. Lang, A. Buchner, G*Power 3: a flexible statistical power analysis program for the social, behavioral, and biomedical sciences, Behav. Res. Methods 39 (2) (2007) 175–191, https://doi.org/10.3758/BF03193146.
[85] Hair, J., Hult, G. T. M., Ringle, C., Sarstedt, M. (2017). *A Primer on Partial Least Squares Structural Equation Modeling (PLS-SEM)* (2nd ed.).
[86] J.M. Roos, A. Kazemi, The five factor model of personality as predictor of online shopping: Analyzing data from a large representative sample of Swedish internet users, Cogent Psychol. 9 (1) (2022), https://doi.org/10.1080/23311908.2021.2024640.
[87] A. Hermes, C. Sindermann, C. Montag, R. Riedl, Exploring online and in-store purchase willingness: associations with the big five personality traits, trust, and need for touch, Front. Psychol. 13 (2022), https://doi.org/10.3389/fpsyg.2022.808500.
[88] N.A. Dodoo, C.M. Padovano, Personality-based engagement: an examination of personality and message factors on consumer responses to social media advertisements, J. Promot. Manag. 26 (4) (2020) 481–503, https://doi.org/10.1080/10496491.2020.1719954.
[89] R. Orji, L. Nacke, C. Di Marco, Towards Personality-Driven Persuasive Health Games and Gamified Systems. (2017), https://doi.org/10.1145/3025453.30255.
[90] A.C.T. Klock, I. Gasparini, M.S. Pimenta, J. Hamari, Tailored gamification: a review of literature, Int. J. Hum Comput Stud. 144 (2020) 102495, https://doi.org/10.1016/j.ijhcs.2020.102495.
[91] G. Tondello, A. Mora, L. Nacke, Elements of Gameful Design Emerging from User Preferences. (2017), https://doi.org/10.1145/3116595.3116627.
[92] M. Denden, A. Tlili, F. Essalmi, M. Jemni, N.-S. Chen, D. Burgos, Effects of gender and personality differences on students' perception of game design elements in educational gamification, Int. J. Hum Comput Stud. 154 (2021) 102674, https://doi.org/10.1016/j.ijhcs.2021.102674.
[93] M. Denden, A. Tlili, S. Salha, M. Abed, Opening up the gamification black box: effects of students' personality traits and perception of game elements on their engaged behaviors in a gamified course, Technol. Knowl. Learn. 29 (2) (2024) 921–940, https://doi.org/10.1007/s10758-023-09701-6.
[94] J. Ul Islam, Z. Rahman, L.D. Hollebeek, Personality factors as predictors of online consumer engagement: an empirical investigation, Mark. Intell. Plan. 35 (4) (2017) 510–528, https://doi.org/10.1108/MIP-10-2016-0193.
[95] G. Dash, J. Paul, CB-SEM vs PLS-SEM methods for research in social sciences and technology forecasting, Technol. Forecast. Soc. Chang. 173 (August) (2021) 121092, https://doi.org/10.1016/j.techfore.2021.121092.
[96] Ringle, Christian M. and Wende, Sven and Becker, J.-M. (2024). SmartPLS4. In *SmartPLS*. https://www.smartpls.com/.
[97] J. Hair, A. Alamer, Partial least squares structural equation modeling (PLS-SEM) in second language and education research: guidelines using an applied example, Research Methods in Applied Linguistics 1 (3) (2022) 100027, https://doi.org/10.1016/j.rmal.2022.100027.
[98] M. Sarstedt, J.F. Hair Jr, J.H. Cheah, J.M. Becker, C.M. Ringle, How to specify, estimate, and validate higher-order constructs in PLS-SEM, Australas. Mark. J. 27 (3) (2019) 197–211, https://doi.org/10.1016/j.ausmj.2019.05.003.
[99] A. Diamantopoulos, P. Riefler, K. Zeugner-Roth, Advancing formative measurement models, J. Bus. Res. 61 (2008) 1203–1218, https://doi.org/10.1016/j.jbusres.2008.01.009.
[100] F. Hair Jr, J., Sarstedt, M., Hopkins, L., & G. Kuppelwieser, V. (2014). Partial least squares structural equation modeling (PLS-SEM). *European Business Review*, *26*(2), 106–121. 10.1108/EBR-10-2013-0128.
[101] Hair, J., Hult, G. T. M., Ringle, C., & Sarstedt, M. (2014). *A Primer on Partial Least Squares Structural Equation Modeling.*

[102] C.B. Jarvis, S.B. MacKenzie, P.M. Podsakoff, A critical review of construct indicators and measurement model misspecification in marketing and consumer research, J. Consum. Res. 30 (2) (2003) 199–218, https://doi.org/10.1086/376806.

[103] J.-M. Becker, K. Klein, M. Wetzels, Hierarchical latent variable models in PLS-SEM: guidelines for using reflective-formative type models, Long Range Plan. 45 (2012) 359–394, https://doi.org/10.1016/j.lrp.2012.10.001.

[104] J.F. Hair, C.M. Ringle, M. Sarstedt, PLS-SEM: indeed a silver bullet, J. Mark. Theory Pract. 19 (2) (2011) 139–152, https://doi.org/10.2753/MTP1069-6679190202.

[105] Wilson, B., & Henseler, J. (2007). *Modeling Reflective Higher-Order Constructs using Three Approaches with PLS Path Modeling: A Monte Carlo Comparison* (pp. 791–800). ANZMAC. https://research.utwente.nl/en/publications/modeling-reflective-higher-order-constructs-using-three-approache.

[106] H.H. Harman, *Modern factor analysis*, University of Chicago Press, 1976 https://books.google.co.in/books?id=e-vMN68C3M4C.

[107] K.A. Bollen, *Structural Equations with Latent Variables*, Wiley (1989), https://doi.org/10.1002/9781118619179.

[108] J. Hulland, Use of partial least squares (PLS) in strategic management research: a review of four recent studies, Strateg. Manag. J. 20 (2) (1999) 195–204, https://doi.org/10.1002/(SICI)1097-0266(199902)20:2<195::AID-SMJ13>3.0.CO;2-7.

[109] R.P. Bagozzi, Y. Yi, On the evaluation of structural equation models, J. Acad. Mark. Sci. 16 (1) (1988) 74–94, https://doi.org/10.1007/BF02723327/METRICS.

[110] J.F. Hair, R.E. Anderson, R.L. Tatham, *Multivariate data analysis with readings*, 2nd ed., Macmillan Publishing Co. Inc., 1986.

[111] Nunnally, J. C. (1978). *Psychometric Theory*. McGraw-Hill. https://books.google.co.in/books?id=WE59AAAAMAAJ.

[112] W.W. Chin, A. Gopal, Adoption intention in GSS, ACM SIGMIS Database: the DATABASE for Advances in Information Systems 26 (2–3) (1995) 42–64, https://doi.org/10.1145/217278.217285.

[113] C. Fornell, D.F. Larcker, Evaluating structural equation models with unobservable variables and measurement error, J. Mark. Res. 18 (1) (1981) 39, https://doi.org/10.2307/3151312.

[114] J. Henseler, C.M. Ringle, M. Sarstedt, A new criterion for assessing discriminant validity in variance-based structural equation modeling, J. Acad. Mark. Sci. 43 (1) (2015) 115–135, https://doi.org/10.1007/s11747-014-0403-8.

[115] J.F. Hair, G.T.M. Hult, C.M. Ringle, M. Sarstedt, N.P. Danks, S. Ray, Partial least squares structural equation modeling (PLS-SEM) using R: a workbook, Business. Springer International Publishing, In Classroom Companion, 2021 https://link.springer.com/10.1007/978-3-030-80519-7.

[116] A. Gold, A. Malhotra, A. Segars, Knowledge management: an organizational capabilities perspective, J. Manag. Information Syst. 18 (2001) 185–214.

[117] J.F. Hair, C.M. Ringle, M. Sarstedt, Partial least squares: the better approach to structural equation modeling? Long Range Plan. 45 (5–6) (2012) 312–319, https://doi.org/10.1016/j.lrp.2012.09.011.

[118] J.F. Hair, M. Sarstedt, C.M. Ringle, J.A. Mena, An assessment of the use of partial least squares structural equation modeling in marketing research, J. Acad. Mark. Sci. 40 (3) (2012) 414–433, https://doi.org/10.1007/s11747-011-0261-6.

[119] J.F. Hair, C.M. Ringle, M. Sarstedt, Partial least squares structural equation modeling: rigorous applications, better results and higher acceptance, Long Range Plan. 46 (1–2) (2013) 1–12, https://doi.org/10.1016/j.lrp.2013.01.001.

[120] W.W. Chin, The partial least squares approach for structural equation modeling, in: *Modern Methods for Business Research*, Lawrence Erlbaum Associates Publishers, 1998, pp. 295–336.

[121] S. Geisser, A predictive approach to the random effect model, Biometrika 61 (1) (1974) 101–107, https://doi.org/10.1093/biomet/61.1.101.

[122] S. Streukens, S. Leroi-Werelds, Bootstrapping and PLS-SEM: a step-by-step guide to get more out of your bootstrap results, Eur. Manag. J. 34 (6) (2016) 618–632, https://doi.org/10.1016/j.emj.2016.06.003.

[123] J.M. Becker, J.H. Cheah, R. Gholamzade, C.M. Ringle, M. Sarstedt, PLS-SEM's most wanted guidance, Int. J. Contemp. Hosp. Manag. 35 (1) (2023) 321–346, https://doi.org/10.1108/IJCHM-04-2022-0474.

[124] Huizinga, J. T. A.-T. T.-. (1950). *Homo ludens : a study of the play-element in culture* (First Beac). Beacon Press. LK - https://worldcat.org/title/600592502.